SONiX 32-Bit Cortex-M0 Micro-Controller

SysTick Example Code

AMENDMENT HISTORY

Version	Date	Description
1.0	2024/12/03	1. First version.

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1 OVERVIEW

This document provides our customers the C program examples of SysTick.

2 FEATURE

- 24-bit SysTick Timer
- Interrupt or polling mode
- Use ARM-compiler V6.12

3 HW

- The starter kit board of vendor desired MCU.

4 EXAMPLE CODE

4.1 Polling Flags

1. Open SysTick.h, and set SYSTICK_IRQ as POLLING_METHOD.

```
/*_____ D E F I N I T I O N S ______________________________________________*/
#define SYSTICK_IRQ POLLING_METHOD      //INTERRUPT_METHOD: Enable SysTick timer and interrupt
                                        //POLLING_METHOD:   Enable SysTick timer ONLY

/*_____ M A C R O S ________________________________________________________*/
```

2. Call SysTick_Init();

```
int   main (void)
{
      //initSysTick
      SysTick_Init();

      .
      .
      User code
      .
      .
}
```

3. Modify the number inside the following red circle to set the desired interval time (unit: ms) in SysTick_Init().

   Example: Interval Time = 10ms

```
/*****************************************************************************
* Function    : SysTick_Init
* Description : Initialization of SysTick timer
* Input       : None
* Output      : None
* Return      : None
* Note        : None
*****************************************************************************/
void  SysTick_Init (void)
{
  SystemCoreClockUpdate();

  __SYSTICK_SET_TIMER_PERIOD(10);     //RELOAD = (system tick clock frequency × 10 ms)/1000 -1

  __SYSTICK_CLEAR_COUNTER_AND_FLAG;

#if SYSTICK_IRQ == INTERRUPT_METHOD
  SysTick->CTRL = 0x7;        //Enable SysTick timer and interrupt
#else
  SysTick->CTRL = 0x5;        //Enable SysTick timer ONLY
#endif
}
```

4.2 Interrupt

1. Open SysTick.h, and set SYSTICK_IRQ as INTERRUPT_METHOD

```
/*_____ D E F I N I T I O N S ______________________________________________*/
#define SYSTICK_IRQ INTERRUPT_METHOD    //INTERRUPT_METHOD: Enable SysTick timer and interrupt
                                        //POLLING_METHOD:   Enable SysTick timer ONLY

/*_____ M A C R O S ________________________________________________________*/
```

2. Call SysTick_Init();

```
int   main (void)
{
      //initSysTick
      SysTick_Init();

      .
      .
      User code
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}
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3. Modify the number inside the following red circle to set the desired interval time (unit: ms) in SysTick_Init().
   Example: Interval Time = 10ms

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 * Function    : SysTick_Init
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 *****************************************************************************/
void  SysTick_Init (void)
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  SystemCoreClockUpdate();

  __SYSTICK_SET_TIMER_PERIOD(10);     //RELOAD = (system tick clock frequency × 10 ms)/1000 -1

  __SYSTICK_CLEAR_COUNTER_AND_FLAG;

#if SYSTICK_IRQ == INTERRUPT_METHOD
  SysTick->CTRL = 0x7;      //Enable SysTick timer and interrupt
#else
  SysTick->CTRL = 0x5;      //Enable SysTick timer ONLY
#endif
}
```

5 RESULT

5.1 Polling Flags

The program will stop at the break point in main loop every 10ms.

```
int main (void)
{
  //User can configure System Clock with Configuration Wizard in system_SN32F760.c
  SystemInit();

  PFPA_Init();                          //User shall set PFPA if used, do NOT remove!!!

  //----------------------------------------------------------------------------
  //User Code starts HERE!!!

  SysTick_Init(); //init SysTick

  while (1)
  {
    #if SYSTICK_IRQ == POLLING_METHOD
    if (SysTick->CTRL & SysTick_CTRL_COUNTFLAG_Msk)
    {
        __SYSTICK_CLEAR_COUNTER_AND_FLAG;
    }
    #endif
  }
}
```

5.2 Interrupt

The program will stop at the break point in SysTick_Handler every 10ms.

```
/*****************************************************************************
* Function    : SysTick_Handler
* Description : ISR of SysTick interrupt
* Input       : None
* Output      : None
* Return      : None
* Note        : None
*****************************************************************************/
__irq void SysTick_Handler(void)
{
  __SYSTICK_CLEAR_COUNTER_AND_FLAG;
}
```

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